

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSEAI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Architectural Modelling (CO2)

Group No : 9

Team Members :

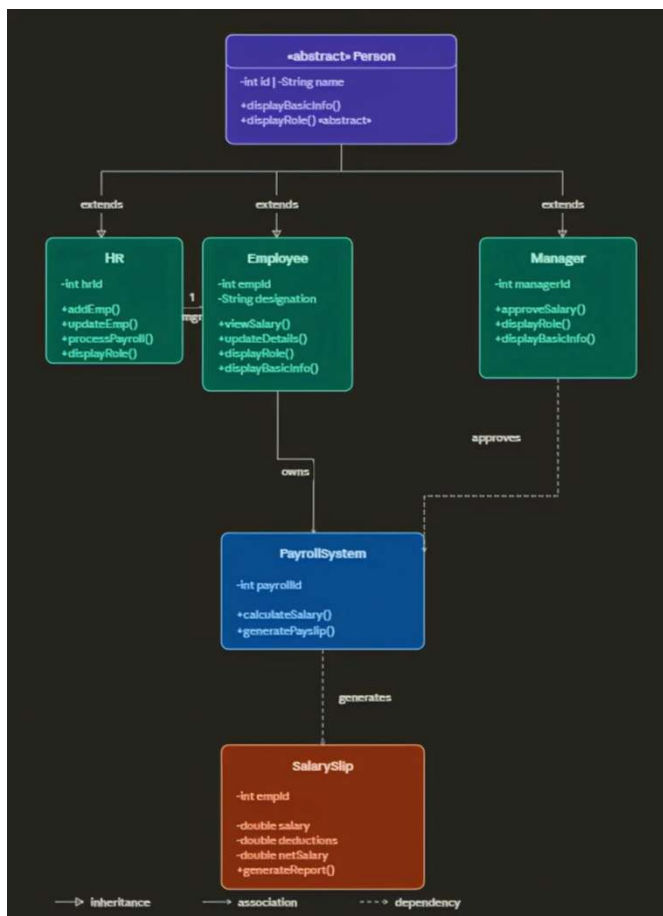
Roll No	Student Name
AM.SC.U4AIE25042	Ananyaa S Pillai
AM.SC.U4AIE25033	Aiswarya Tejas
AM.SC.U4AIE25045	Avanthika Binu
AM.SC.U4AIE25030	Achyuth Narayanan VK

1. Problem Title : Employee payroll management system

STRUCTURAL DESIGN

3. Class diagram

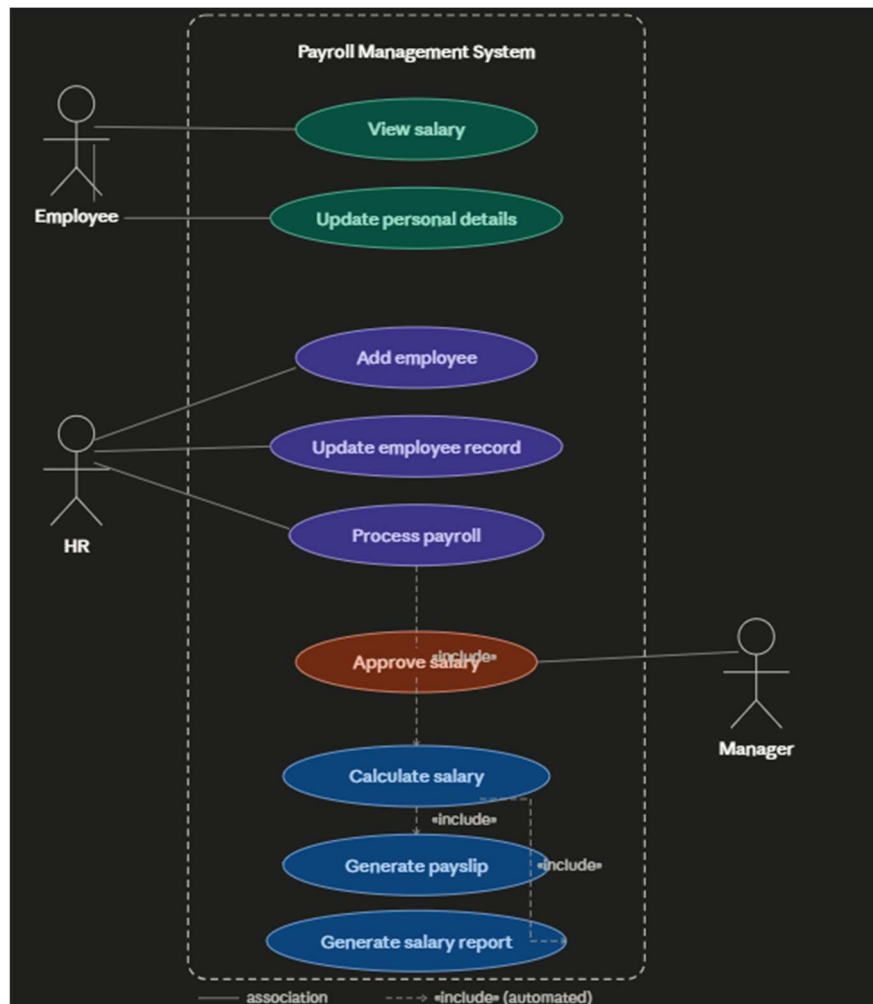
The class diagram below illustrates the main classes in the Employee Payroll Management System along with their attributes, methods, and relationships, drawn using UML notations. Five classes are identified: **Employee**, **HR**, **Manager**, **PayrollSystem**, and **SalarySlip**.



BEHAVIUIORAL DESIGN

4. Use-case diagram

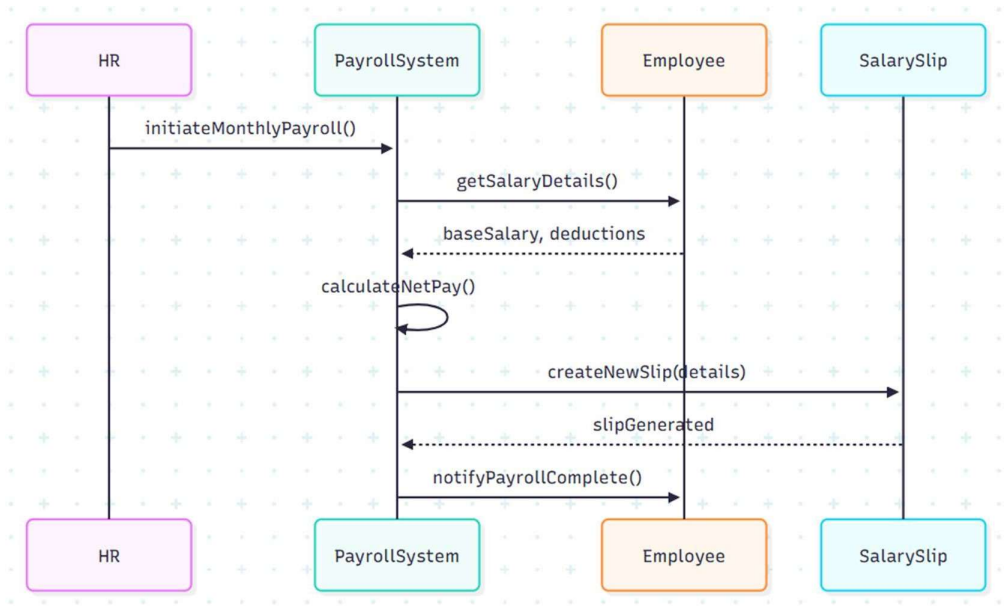
- a. List of actors
 1. **Employee:** A staff member who views payslips and updates personal profiles.
 2. **HR :** An administrator responsible for adding new employees and managing records.
 3. **Manager:** A supervisor who approves attendance and department-level data.
 4. **PayrollSystem:** The automated component that calculates net pay and tax deductions.
- b. List of use-cases
 1. **View Salary Slip:** The employee accesses the system to view or download monthly pay details.
 2. **Manage Employee Records:** HR updates employee salary grades and contact information.
 3. **Approve Timesheets:** Managers verify work hours before the payroll cycle begins.
 4. **Calculate Monthly Salary:** The system processes gross pay and deductions in real-time.
 5. **Process Payment:** The system triggers the final fund transfer to employee accounts.
- c. Use case diagram (use UML notations)



5. Sequence diagram (use UML notations)

a. Scenario: Monthly Salary Generation

The HR administrator initiates the monthly payroll process. The PayrollSystem retrieves salary details and attendance records for each Employee. It calculates the net pay, generates a SalarySlip, and saves the record to the database. Finally, a notification is sent to the employee



b. Scenario: Employee Attendance Logging

An employee logs into the system to request a specific monthly slip. The PayrollSystem queries the database to retrieve the relevant record. Once validated, the system formats the data and displays the SalarySlip to the employee on the portal.

